

DTSA 5701

Introduction to High Performance and Parallel Computing

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (73.3 / 100)	61.2 / 100
SPECIALIZATION Introduction to High Performance and Parallel Computing		

61	98	+50
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

This course introduces the fundamentals of high-performance and parallel computing, and the software skills necessary for work in parallel software environments. These skills include big-data analysis, machine learning, parallel programming, and optimization. It covers the basics of Linux environments and bash scripting all the way to high throughput computing and parallelizing code.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	73.3	100.0	+26.7	MEDIUM
AI Engineer	68.8	100.0	+31.2	HIGH
MLOps Engineer	65.8	100.0	+34.2	HIGH
ML Engineer	57.4	89.6	+32.2	HIGH
Data Scientist	52.6	98.3	+45.8	HIGH
Applied Scientist	49.5	100.0	+50.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Model Deployment & Serving 2 of 6 roles

production

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

- 2 **MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

- 3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

73.3 → 100.0 (+26.7 pts)

data engineering sets analytics tools solutions business mobile

AI Engineer

68.8 → 100.0 (+31.2 pts)

data engineering build technical systems problems technology solutions

MLOps Engineer

65.8 → 100.0 (+34.2 pts)

data production scientists systems engineering tools error

ML Engineer

57.4 → 89.6 (+32.2 pts)

discovery data systems production engineering starting world build

Data Scientist

52.6 → 98.3 (+45.8 pts)

data sets characteristics analytics evolution different analytical social

Applied Scientist

49.5 → 100.0 (+50.5 pts)

lab discovery scientific computational research literature problems technical

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.